

Problem

Linear Superposition Assumption

Activations = sparse feature combinations:
 $\text{activation} = w_1 * f_1 + w_2 * f_2 + \dots$

Persistent Challenges

- SAEs fail to eliminate polysemanticity
- Pathological reconstruction errors

Hypothesis

Dual Encoding Hypothesis

Networks encode in TWO complementary spaces:

1. Feature Identity Space

What concepts are present

'Paris', 'democracy', 'sentiment'

2. Feature Integration Space

How concepts combine

Methods

Sequential Pipeline

- Train primary SAE on activations
- NFM predicts SAE residuals
- Secondary SAE on NFM embeddings

Joint Training (Architecture below)

Key Definitions

Dictionary Feature

Concept identity: $f = a_1 + a_2 + a_3 \dots$

'surprise' detected

Co-activation

Statistical correlation: $\text{corr}(f_1, f_2)$

'surprise' + 'unknown' co-occur

Integration Feature

Computational combination: $I = f_1 + f_2$

'happy' + 'birthday' → JOY

Nonlinear Integration

Nonlinear combination: $I = f_1 * f_2$

'surprise' + 'diagnosis' → DREAD

41.3%

Reconstruction Improvement

51.6%

KL Divergence Reduction

Results

Reconstruction Performance

- Joint: 41.3% vs Sequential: 23.4%
- Linear: 95.5% | Nonlinear: 4.5%

Behavioral (KL Divergence)

- Linear: 50.9% | Nonlinear: 8.6%
- Combined: 51.6% reduction

Emergent Specialization

Bimodal at norm=0.2 ($r = \pm 0.987$)

Low norm → Integration | High → Identity

Key Equations

NFM Components

Linear: $\text{Sum}_i (w_i * f_i) + b$

Nonlinear: $0.5 * (\text{Sum } v * f)^2 - \text{Sum}(v * f)^2$

Final = SAE + Linear + Nonlinear

Limitations

- Scale: 3B model, 50k features, layer 16
- Integration features opaque to inspection
- WikiText-103 domain only

Future Directions

- Cross-model (GPT, Llama, Mistral)
- Cross-layer (early, middle, late)
- Scaling to larger models

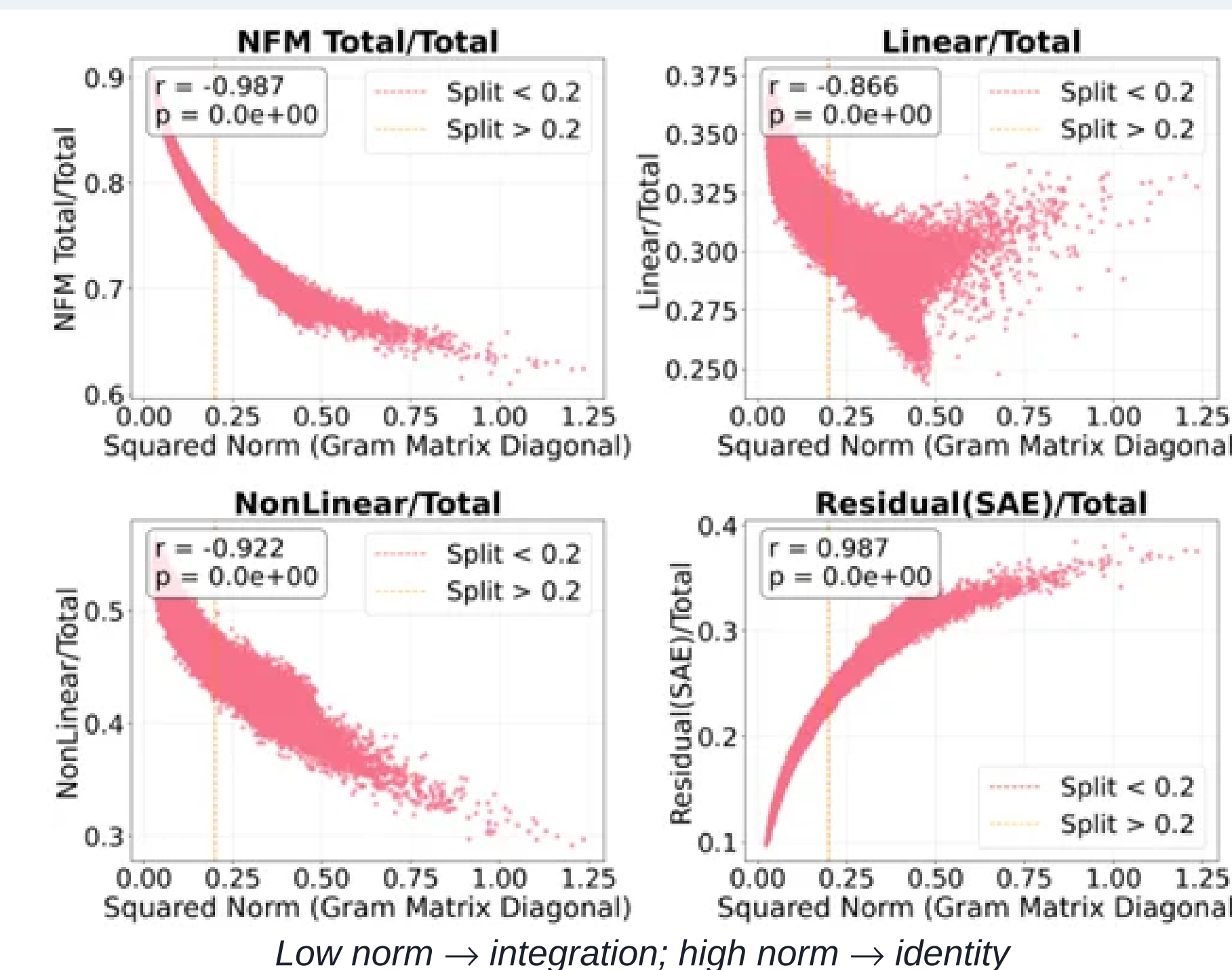
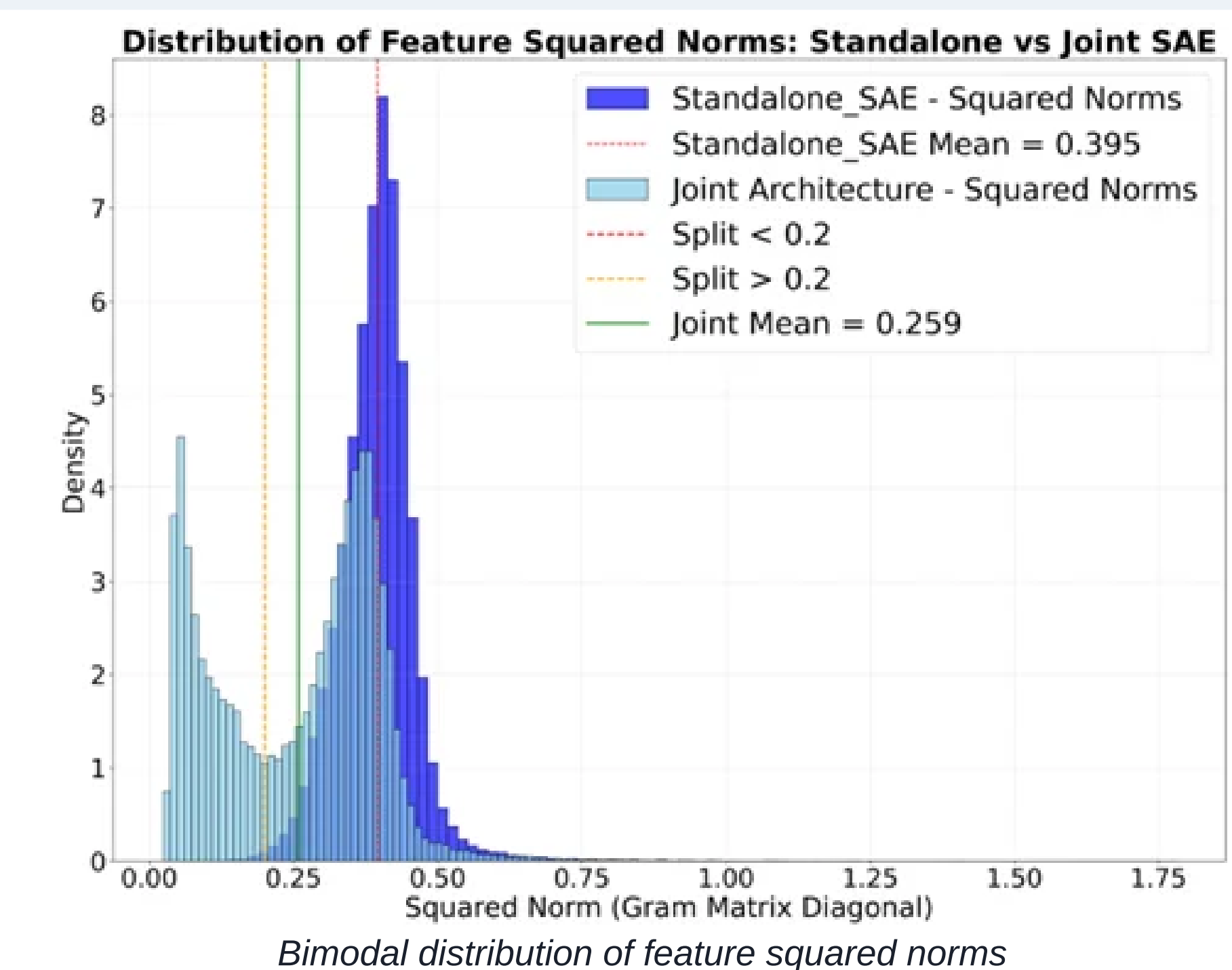
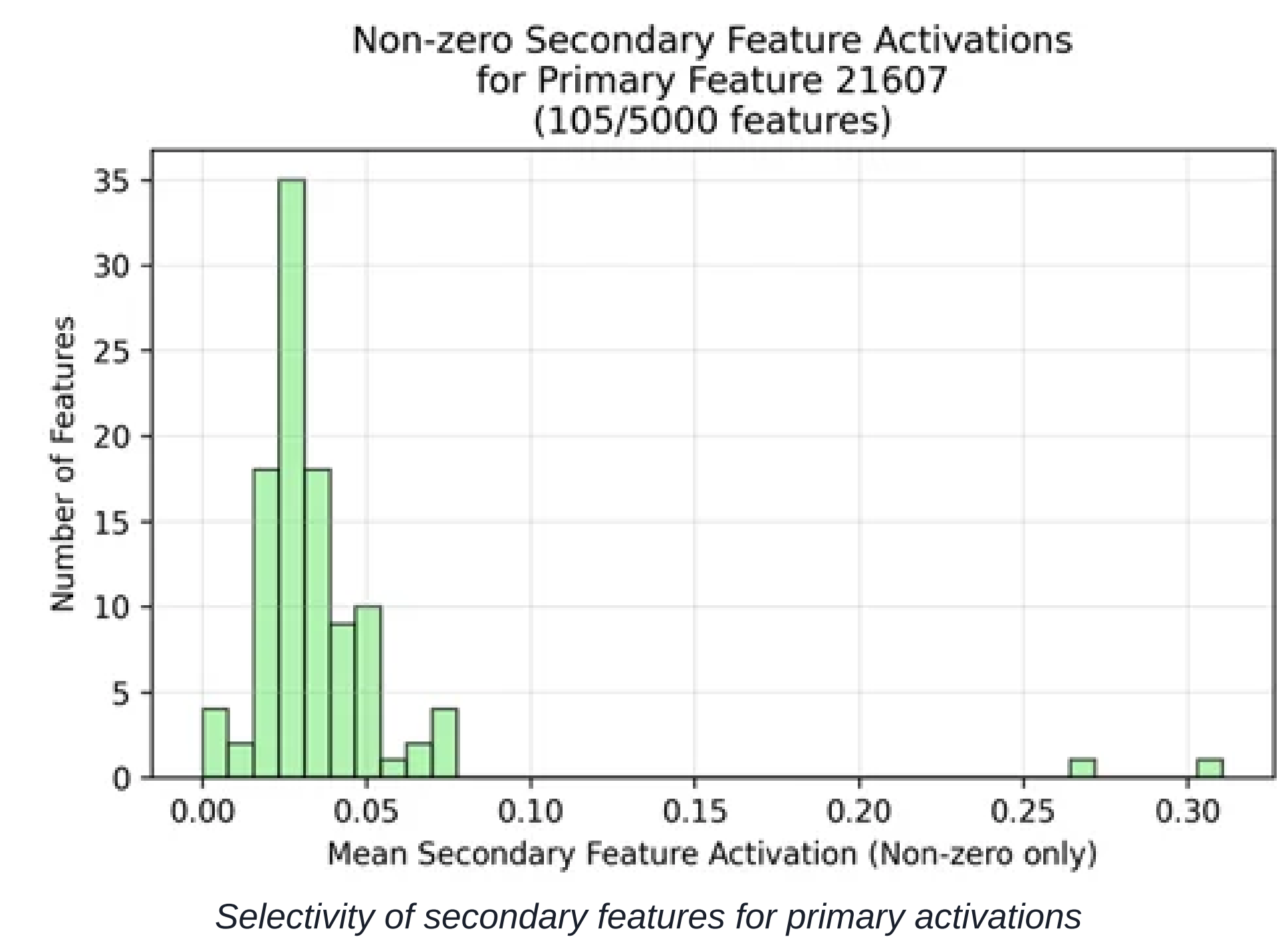
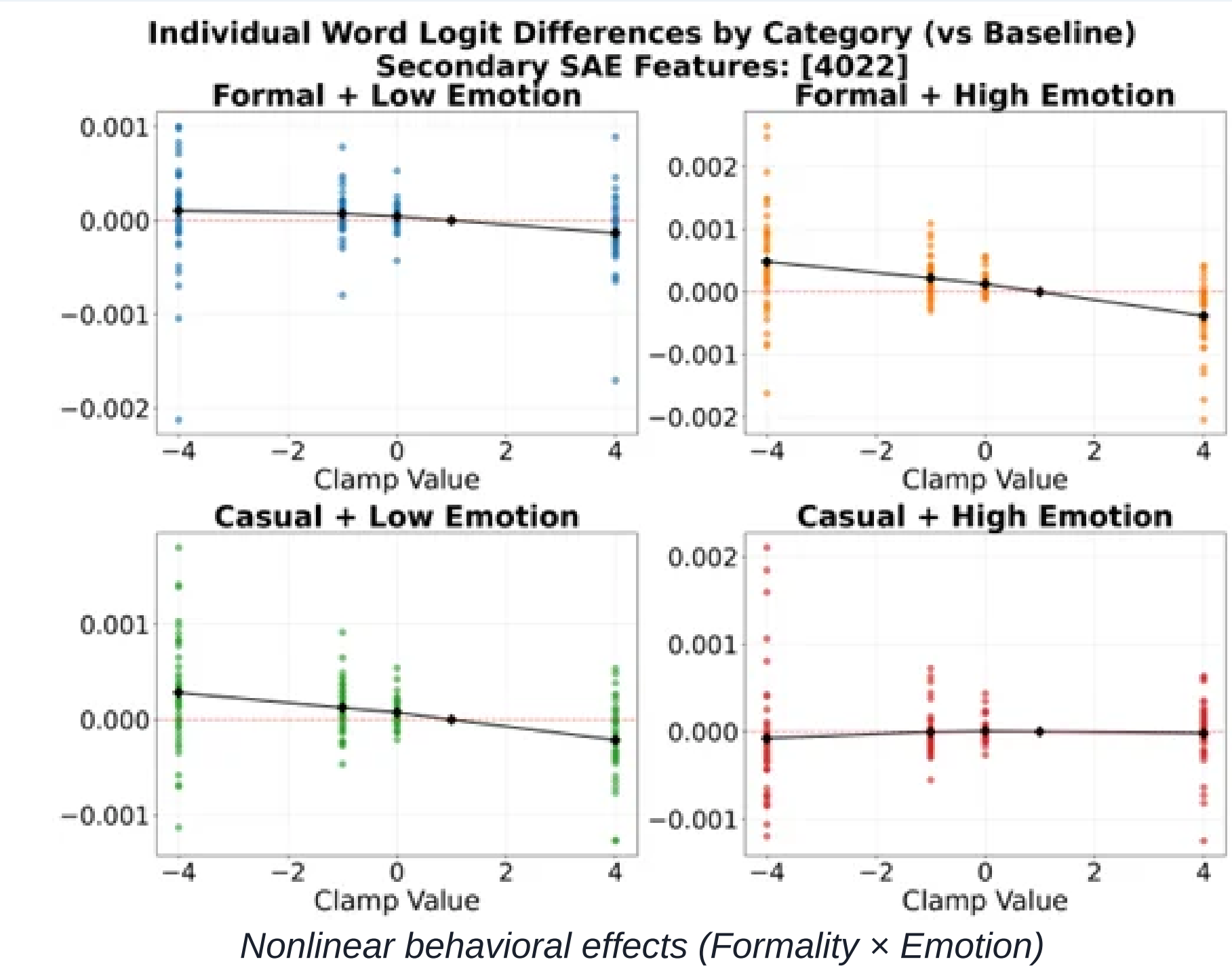
Implications

Theoretical: Polysemantic features may reflect poorly decoded relationships missed by linear decomposition

Practical: Captures missing computation

Safety: Better mechanistic understanding

Paradigm: Post-hoc → Integrated design



Joint Architecture (SAE + NFM)

